Vaibhav Vikas Gaikwad

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SUMMARY

Data Engineer with experience building scalable data pipelines, large-scale ETL systems, and cloud data platforms supporting analytics and machine learning workloads. Proven ability to optimize SQL pipelines, and design data models powering BI dashboards and decision-making systems. Experienced with Spark, Python, SQL, Databricks, and cloud data warehouses to build reliable and production-grade data infrastructure. Passionate about building high-performance, scalable data systems that power real-time analytics and business intelligence.

SKILLS

- **Programming:** Python, SQL, PySpark, R
- **Data Engineering:** ETL Pipelines, Data Modeling, Data Warehousing, Data Quality Validation, Feature Engineering
- **Big Data:** Apache Spark, Kafka, Databricks, Distributed Processing
- **Databases:** PostgreSQL, Azure SQL, Snowflake, dbt
- **Cloud:** AWS, Azure
- **BI & Analytics:** Tableau, Power BI
- **Tools:** Git, Jupyter, Docker, Airflow, Linux, Bash
- **Concepts:** Data Pipelines, CI/CD Pipeline, Workflow Automation, Data Validation, Batch Processing, Analytics Engineering

WORK EXPERIENCE

Data Engineer | American Express, Remote **January 2025 – Present**

- Designed and deployed scalable ETL pipelines to process large-scale financial transaction datasets using Python and SQL, improving data availability by 30% for fraud analytics, risk monitoring, and financial reporting teams.
- Engineered distributed data processing workflows using PySpark on Databricks to handle high-volume payment and card transaction data, reducing data transformation time by 40% for complex financial analytics workloads.
- Implemented automated data quality and validation frameworks to detect schema drift, missing financial records, and transactional anomalies, improving pipeline reliability and maintaining 99% reporting accuracy for financial compliance.
- Developed optimized financial data models and SQL transformations in Snowflake, enabling enterprise-grade BI dashboards for transaction monitoring, customer spending insights, and executive KPI tracking.
- Orchestrated automated data ingestion and transformation pipelines supporting near real-time financial analytics, enabling faster decision-making for fraud detection, credit risk analysis, and customer behavior analytics.
- Built data pipelines supporting fraud detection and risk analytics models, enabling analysis of 10M+ daily transaction events and improving fraud monitoring efficiency by 25%.

Data Engineer | BMR Infotek, India **June 2021 – June 2023**

- Architected and maintained end-to-end ETL pipelines to ingest structured and semi-structured healthcare data (patient records, clinical datasets, and operational metrics) into a centralized PostgreSQL data warehouse, ensuring high data quality and availability for healthcare analytics.
- Processed and transformed 50GB+ of daily healthcare data using Apache Spark, enabling scalable ingestion of electronic health records (EHR), patient monitoring data, and operational datasets for downstream analytics and reporting.
- Developed complex SQL-based healthcare data models and optimized queries to support clinical reporting and operational dashboards, reducing reporting latency by 30% for healthcare administrators, analysts, and operational teams.
- Automated data workflows for healthcare data cleaning, standardization, aggregation, and feature engineering, supporting predictive analytics initiatives such as patient outcome analysis, hospital utilization trends, and care quality monitoring.
- Designed and delivered interactive Tableau dashboards to visualize key healthcare KPIs such as patient admissions, treatment outcomes, and operational efficiency, enabling leadership and clinical stakeholders to make data-driven decisions.
- Implemented data validation and quality monitoring checks across healthcare pipelines to ensure accuracy and compliance with healthcare data governance standards.

PROJECTS

Document AI Data Processing Pipeline (TypeScript, Node.js, Docker, Crossref API):

- Built a scalable document ingestion pipeline that converts scientific PDFs into structured JSON datasets using hybrid parsing, heuristic metadata extraction, and LLM ensemble processing.
- Engineered modular ETL workflows with parser fallbacks (pdf-parse, GROBID, Unstructured) and Crossref API validation, enabling reliable batch document processing and normalized datasets for downstream search and analytics systems.

Clinical Pharmacokinetics Data Platform (Python, PK-DB API, SQLite/PostgreSQL, FastAPI):

- Developed a multi-source data pipeline ingesting pharmacokinetics datasets from PK-DB APIs and molecular ADME benchmarks, processing 160K+ patient records across 700+ studies into a normalized relational schema.
- Implemented automated ETL workflows for data cleaning, feature engineering (BMI, CrCl, BSA), and data validation, generating analytics-ready datasets used for ML modeling, BI dashboards, and real-time prediction APIs.

Real-Time AI Job Market Data Pipeline (Python, Kafka, Spark, Airflow, Parquet, Tableau):

- Built a streaming data pipeline that ingests job posting events for AI/ML and Data Science roles, processes incremental updates through Kafka and Spark Structured Streaming, and transforms raw listings into curated analytics datasets.
- Implemented scheduled orchestration, salary normalization, and skill extraction workflows, storing partitioned Parquet data for downstream dashboards that tracked hiring demand, compensation trends, and geographic shifts across the U.S. labor market.

EDUCATION

Master of Science in Applied Data Science | Syracuse University, School of Information Studies, Syracuse, NY **May 2025**
Bachelor of Engineering | University of Mumbai, India **June 2022**

CERTIFICATIONS

- AWS: Machine Learning Engineer – Associate
- Oracle Cloud: Data Science Associate (OCI)